Games without Chance

Tom Morley

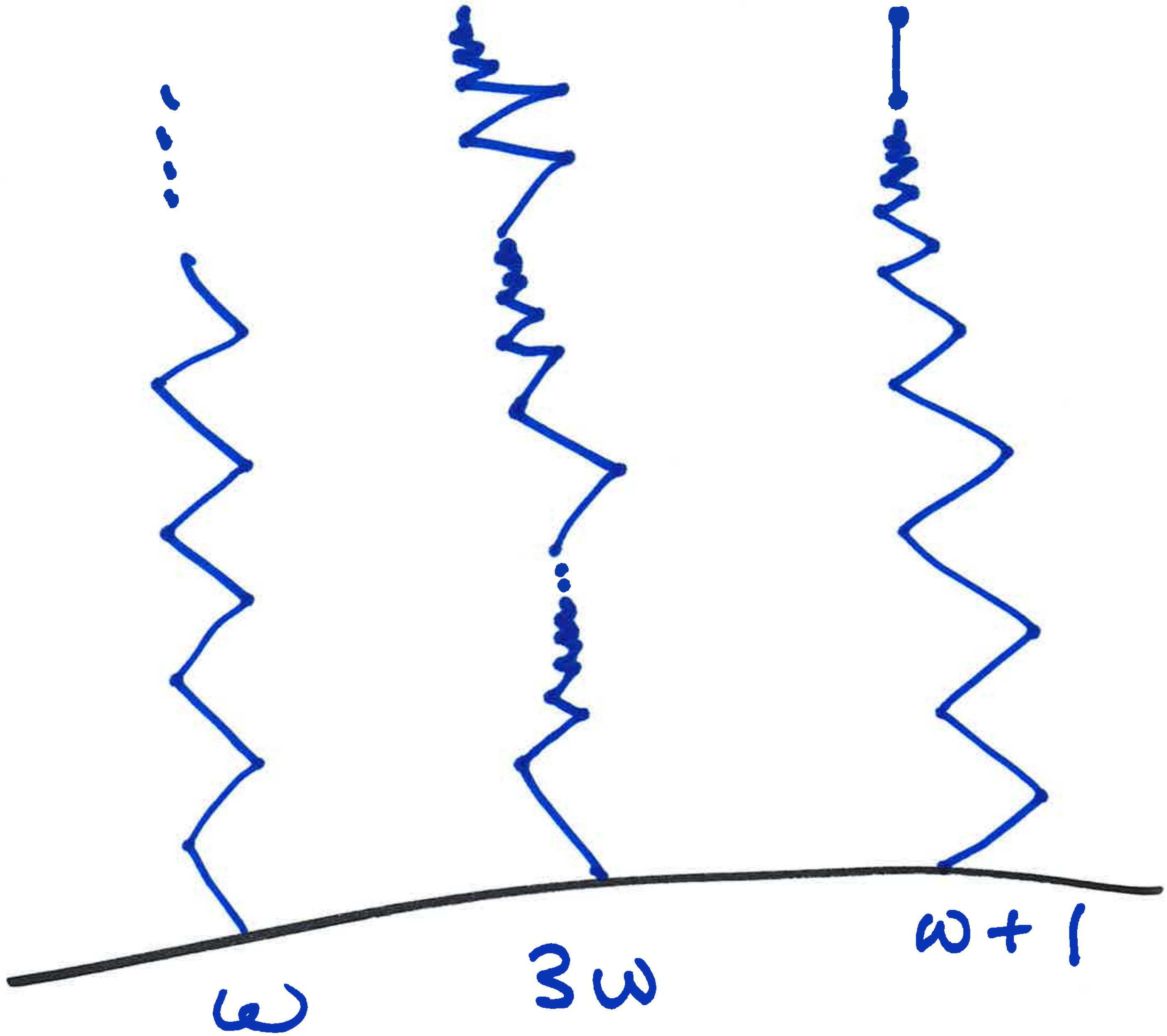
WEEK 7

where do we go from here?

1b.

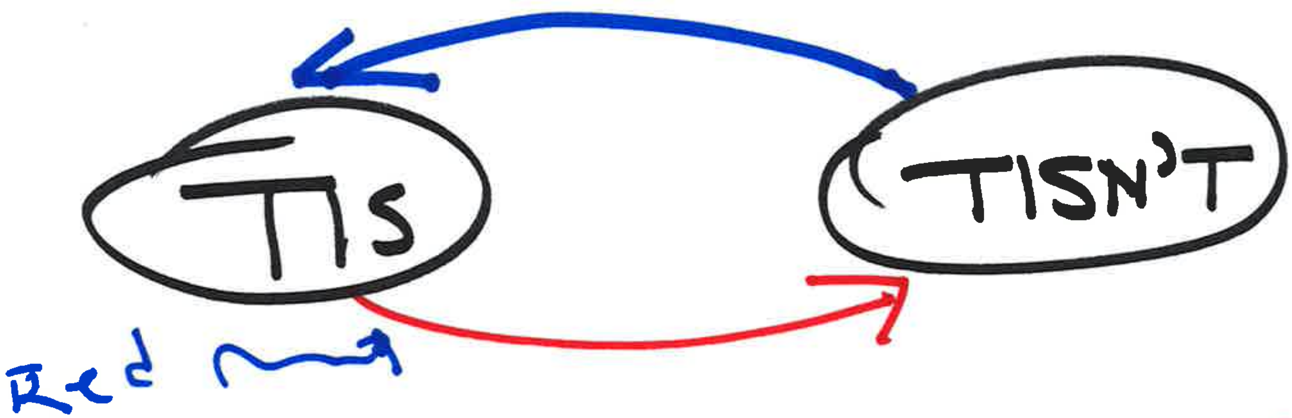
1. Infinite & Loopy games
2. Mean values, hot, cold, sente gote
3. Atomic Weights

...???


ω
3ω
ω+1

Surreal Numbers D. Knuth

includes Real Numbers
(infinite) Ordinal Numbers



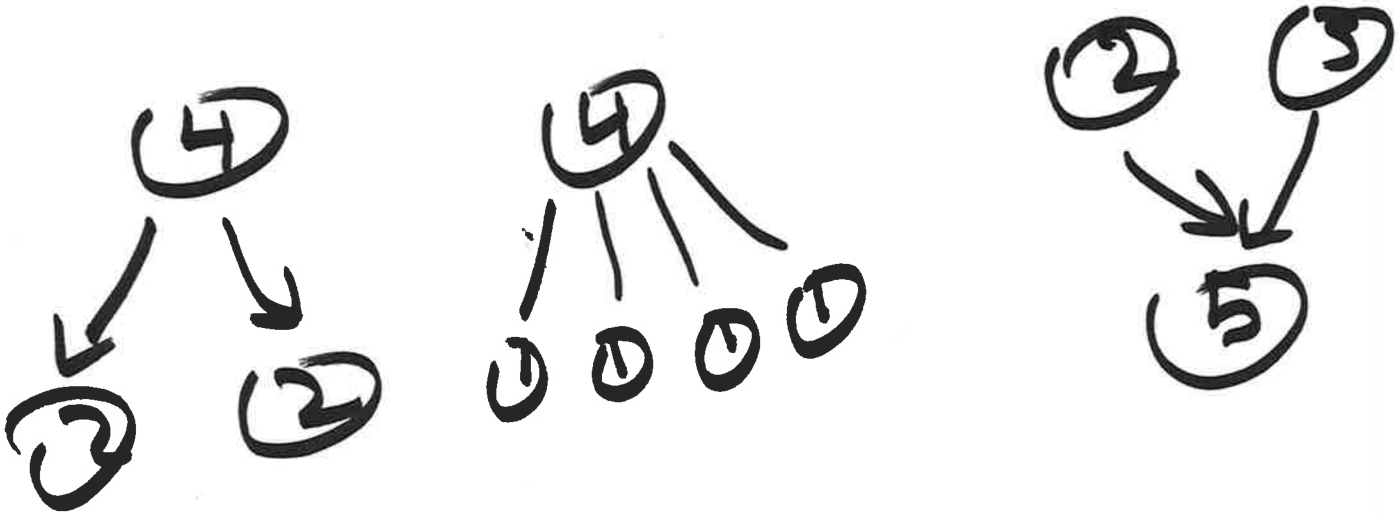
ARGUING IS NOT JUST contradicting...

YES IT IS

NO IT ISNT

yes....

FAIR Shares & Varied pairs



1st PLAYER can't MOVE LOSES...

$\{+500 / -500\}$

G + G + . .

Given G there is a number $m(G)$ with

$$-K \le \underbrace{G + G + \dots + G}_{n \text{ times}} - n \cdot m(G) \le K$$

$-K$ . . independent of n.

Sente

Gote

No one is eager to play $* = \{0|0\}$

But $\int^3 * = \{0+3|0-3\}$

<u>Hot</u> game

Heated infinitesimal

S. NORTON

Any game G is

$$x + \sum \int^{t_i} \varepsilon_i$$

All small $\cancel{\{0|\}}$

If L has a move so does R
if R has a move so does L

TRUE for G and any position

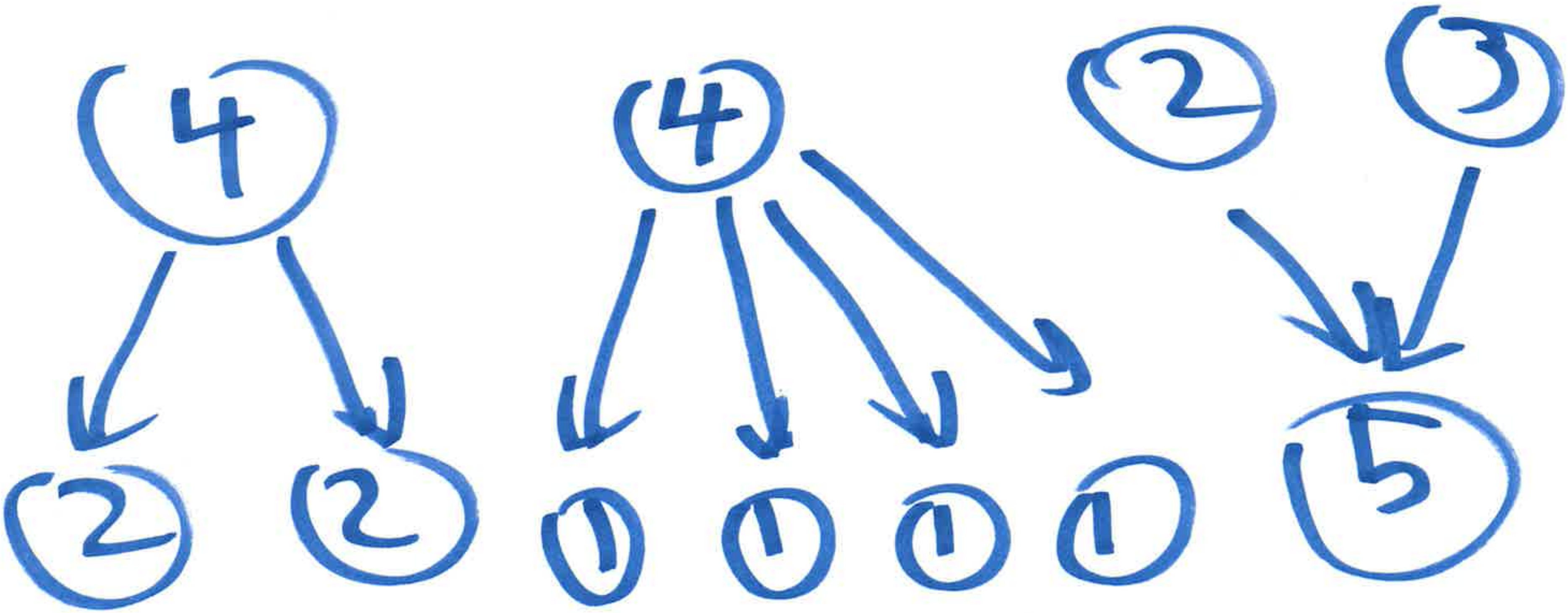
$$-1/2^n < G < 1/2^n$$

Atomic Weights

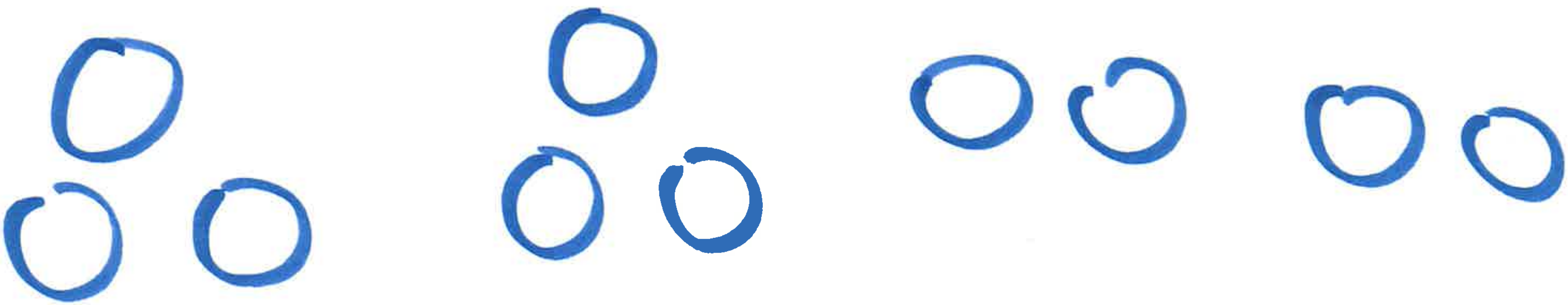
g all small, then there is G, computable from g

$$\downarrow^* + \star \leq g - G.\uparrow \leq \uparrow^* + \star$$

FAIR Shares Varied pairs



Your Move!



3 3 2 2

Let's Play

A game!